

Haar Cascade Face Detection Questions & Answers

Name: Vanshika Bisht
Course: BSc Data Science
College: AAFT

Q: What is Haar Cascade Face Detection?

Haar Cascade is a machine learning-based method used to detect objects like faces in images.

- It is trained using many **positive images (with faces)** and **negative images (without faces)**
- It looks for specific features like **edges, lines, and patterns** in the image
- It uses a cascade of classifiers to quickly detect faces

In simple words:

It is a fast method that scans an image and finds faces by checking patterns.

Q: Why do we convert an image to grayscale before detection?

We convert the image to grayscale because:

- It reduces complexity (only 1 channel instead of 3: RGB)
- Makes processing faster
- Haar Cascade only needs intensity (brightness), not color

Simple words:

Grayscale helps in faster and more efficient face detection.

Q: Python Program for Face Detection.

```
import cv2
```

```
# Load Haar Cascade file
```

```
face_cascade = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')
```

```
# Read image
```

```
img = cv2.imread('image.jpg')
```

```
# Convert to grayscale
```

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
# Detect faces
```

```
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5)
```

Draw rectangle around faces

for (x, y, w, h) in faces:

```
cv2.rectangle(img, (x, y), (x+w, y+h), (255, 0, 0), 2)
```

Show image

```
cv2.imshow('Face Detection', img)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Q: Explain detectMultiScale()

```
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5)
```

This function detects faces in an image.

- **gray** → input grayscale image
- **scaleFactor=1.3** → reduces image size at each step (helps detect different face sizes)
- **minNeighbors=5** → number of nearby detections required to confirm a face

Simple words:

It scans the image at different sizes and finds faces based on confidence.

Q: What does (x, y, w, h) represent?

- **x** → x-coordinate (left position)
- **y** → y-coordinate (top position)
- **w** → width of the face
- **h** → height of the face

What does this loop do?

for (x, y, w, h) in faces:

- It goes through all detected faces
- For each face, it gives position and size

- Used to draw rectangles around faces

Simple words:

Loop is used to process and mark each detected face.

Q: Webcam Face Detection Code.

```
import cv2

face_cascade = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')

# Start webcam
cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    faces = face_cascade.detectMultiScale(gray, 1.3, 5)

    for (x, y, w, h) in faces:

        cv2.rectangle(frame, (x,y), (x+w,y+h), (255,0,0), 2)

    cv2.imshow('Webcam Face Detection', frame)

    # Press 'q' to exit

    if cv2.waitKey(1) & 0xFF == ord('q'):

        break

cap.release()

cv2.destroyAllWindows()
```

Q: What are limitations of Haar Cascade?

- Not very accurate for complex images
- Poor performance in low light
- Struggles with rotated or side faces

- Can give false detections
- Less accurate compared to modern deep learning methods (like CNN)

Simple words:

It is fast but not very accurate in difficult situations.